FLEET TEAM DIGEST

737MAX-FTD-29-22002

Issue Title : Commanded IFSD due to Hydraulic Fluid Contamination of the Engine Oil Resulting From EDP Failure (737NG/MAX)

Airplane Model	ATA	Minor Model(s)
737MAX	2911-11	-7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
737, 737NG	9E991	

Originated Date	Last Revised Date	Created On
07/14/2022	04/23/2025	04/18/2025
Estimated Completion Date	**Next Update Date**	
10/30/2025	09/29/2025	

Revision Description

No new updates to report.

Description

The intent of this article is to inform operators of recent commanded IFSD (inflight shutdown) events due to hydraulic fluid contamination of the engine oil, resulting from EDP (engine driven pump) overheat failure on 737NG and MAX airplanes.

All IFSD events have been associated PN 849589 (one of two approved suppliers). Both approved EDP suppliers have experienced failures associated with overheat events, however contamination of the engine oil system and subsequent commanded IFSD has only been associated to the PN: 849589 supplier.

Multiple 737NG and 737 MAX airplanes have reported these events, a majority have been attributed to maintenance induced errors such as incorrectly coupled quick disconnect or collapsed flexible hose.

EDP cooling is accomplished via an unrestricted exchange of fluid from EDP through the case-drain flow path returning to the aircraft hydraulic system.

Status

Boeing is working in coordination with the supplier of the EDP, Eaton and the engine manufacturer, CFM to investigate the cause of the commanded IFSD events.

The EDP failed due to overheat, Boeing has not established a definitive root cause for the overheat. Boeing is aware of events with known failure modes associated with human factors, associated to hydraulic quick disconnects, and its installation (creating a restriction).

Investigation Findings:

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• Distressed Fluid (evidence of coking indicates temp above 350F), allowing fluid to coke and creating a restriction.

Note: Past 737 NG events, except for one, were associated to an subsequent EDP failure that resulted in Oil filter bypass illumination, and commanded IFSD. Some of these events were associated with human factors. Therefore, Boeing has improved the AMM EDP procedure for both the NG and MAX to mitigate subsequent EDP failure events following an EDP replacement. Boeing also updated the 737 MAX AMM to include revised guidance to not pull the hose QD rotating sleeve when verifying its installation.

Boeing held a workshop in January 2022 to review the investigations findings and discuss a plan forward. Boeing also explored design solutions and identified modification of the AGB pad drain size as mitigating action for commanded IFSDs.

A Multi-Operator Message (MOM) has been released to bring awareness to 737 MAX Quick Disconnect (QD) installation practices and refer operators and MROs to an AMM supplemental video that is now available. Boeing will be releasing a Hydraulic Quick Disconnect FTD article to discuss human factors, its design, and summarize best practices.

A retrofit Service Bulletin has been published ahead of production, with limited parts available starting on February 17. Production incorporation will begin with L/N 9230.

Interim Action

Boeing has updated the 737NG/MAX AMM EDP removal procedures to enhance the warning and caution notes to help mitigate maintenance induced EDP events.

Boeing is updating the 737MAX AMM to provide detailed hydraulic quick-disconnect installation verification checks (including reminders not to pull the rotating sleeve (axially) - to not inadvertently pull the locking tab out of the slot of the hoses).

Boeing has updated the 737NG/MAX AMM to note it is not necessary to remove the case drain filter hose at the strut during a case drain filter change.

Additionally, Boeing is assessing potential design solutions to mitigate IFSDs (in absence of definitive root cause) and Boeing has added another hydraulic quick-disconnect inspection prior to new airplane deliveries.

Options being evaluated include:

- (For 737 MAX Only) EDP Pad Drain System: Design change to ensure Pad Drain Line tolerates predicted flow from failed EDP and prevents ingress of hydraulic fluid in AGB

If operators experience a commanded IFSD caused by an EDP failure, Boeing recommends contacting Boeing and CFM for further instructions.

However, with the Oil Filter Approaching or Impending bypass message, operators can take the following mitigating maintenance actions:

- Check for EDP shaft seal leakage
- Check hydraulic fluid and engine oil quantities for a transfer of fluid
- Check/replace the EDP case drain filter (note the condition when removed)

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- Check for proper QD lock tab installation at pump case, pump inlet, and pylon connections
- Inspect for debris in case drain filter after replacement
- Flush the hydraulic system, inspect EDP case drain lines
- Check/replace the AGB rotating seal
- Visually inspect gearbox drain for obstructions
- Replace the oil filter
- Flush the engine oil
- Check the case drain filter module and associated check valve for free flow arrow direction (pointing AFT)
- Confirm case drain filter module check valve is free from blockage/debris

Report any inspection findings to Boeing/CFM/Supplier.

For airplanes in storage, Boeing recommends -

- Ensure case drain quick-release fitting and hydraulic line hoses are properly installed
- EDP case drain filter element inspections per 737MAX AMM TASK 29-11-51-200-801

Final Action

The root cause to EDP pump overheat events that are not related to improper maintenance are unknown at this time. Boeing is moving forward with the 737 MAX AGB drain line resizing to accommodate a greater hydraulic fluid flow.

The service bulletin SB-737-71-2013 has been published with instructions to remove and install new drain tube assemblies connected to the pad drain of the EDP, Integrated Drive Generator (IDG), and hand crank.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: Complete

Solution Selected: Complete

737MAX AMMs: Temporary Revision (TR) to the 737 MAX AMM to include revised guidance to not pull the hose QD rotating sleeve when verifying its installation - Complete

737MAX QD Verification Multi-Operator Message (MOM), MOM-MOM-24-0013: Complete

737 MAX Pad Drain Line Change Committed: Complete

737 MAX Retrofit Service Bulletin: SB-737-71-2013 R00 released on January 22, 2025 (Complete)

737 MAX Parts Available: February 17, 2025 (limited parts availability, part allocation/control)

737 MAX Production Incorporation: September 2025 (L/N 9230) (Target)

Reference Categories

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Reference Type	Reference Number	Revision Number	Reference Date
Multi-Operator Message (MOM)	MOM-MOM-24-0013-01B		
Service Bulletin (SB)	SB-737-71-2013	R00	01/22/2025

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Engine Driven Pump (EDP)	849589		
Top Kit - EDP Pad Drain Line Replacement	002A0140-1	Boeing	

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- AOGs : Events that cause a significant Airplane On Ground (AOG) condition
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations
- Fleet Team IdeaXchange : Topic of discussion on the Fleet Team IdeaXchange bulletin board
- New Equipment : Information on new parts available now or soon to be introduced
- Significant Event Report : List of significant in-service or maintenance events reported to Boeing that result in
  Air turnbacks, ground turnbacks, diversions, rejected takeoffs, in-flight shutdowns, or Significant AOGs.
  High failure rate of costly/critical systems
- Significant Technical Flight Interruptions : Issues that cause in-flight events such as rejected takeoffs, air turnbacks, diversions, emergency landings, in-flight shutdowns, and abnormal stops

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